NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

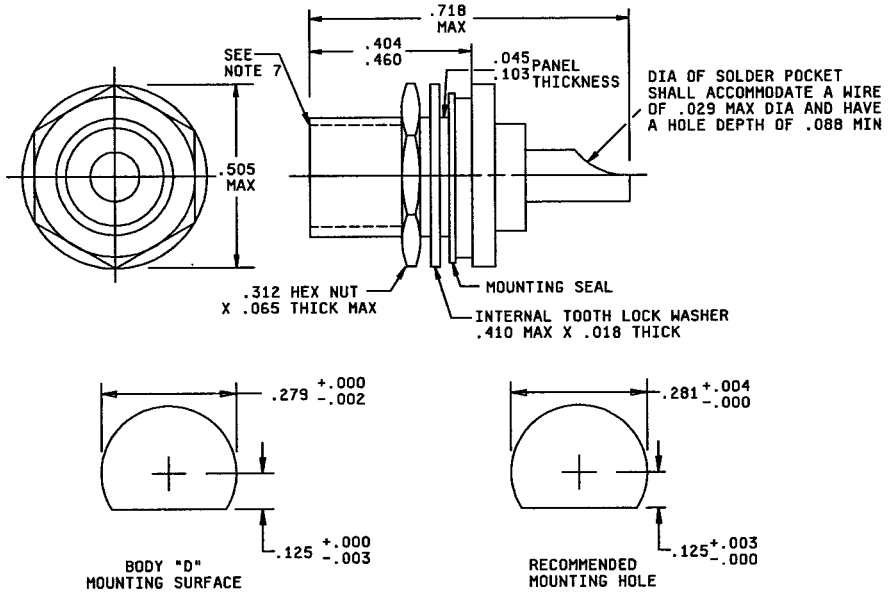
MIL-PRF-39012/61B
25 January 1995
SUPERSEDING
MIL-C-39012/61A
1 September 1984

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA
(UNCABLED, SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments
And Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-39012



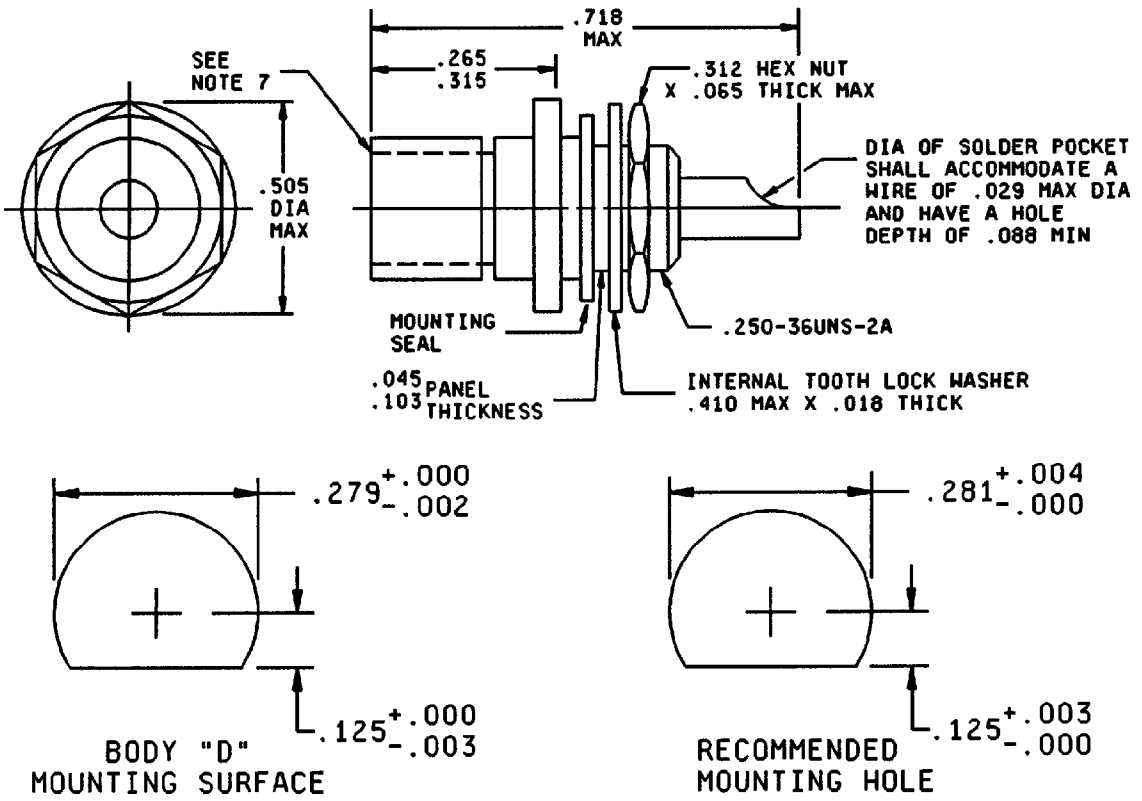
REAR MOUNT (M39012/61-3001 AND M39012/61-4001)

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.088	2.24	.312	7.92
.003	0.08	.103	2.62	.315	8.00
.004	0.10	.125	3.18	.404	10.26
.018	0.46	.250	6.35	.410	10.41
.029	0.74	.265	6.73	.460	11.68
.045	1.14	.279	7.09	.505	12.83
.065	1.65	.281	7.14	.718	18.24

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:01Z
Check the source to verify that this is the current version before use.



FRONT MOUNT (M39012/61-3002 AND M39012/61-4002)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension .505 inch (12.83 mm) is the largest overall diameter of the connector.
4. All undimensioned pictorial configurations are for reference purposes only.
5. There shall be a solid barrier in the socket between the pin entry and the solder pocket to prevent solder wicking.
6. Solder pockets shall be gold plated in accordance with MIL-G-45204, type II, class 1.
7. Series SMA socket contact interface shall be in accordance with MIL-STD-348.

FIGURE 1. General configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:01Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: Not rated.

Voltage rating: 335 volts rms maximum at sea level; 85 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics: See MIL-STD-348 for dimensions.

Contacts with spring members:

Center contact (female):

Oversize test pin: .0375 +.0001/-.0000 inch.

Test pin finish: 16 microinches.

Insertion depth: .030/.045 inch.

Number of insertions: 3.

Insertion force test:

Steel test pin diameter: .0370 +.0001/-.0000 inch.

Insertion depth: .050/.075 inch.

Test pin finish: 16 microinches.

Insertion force: 2 pounds maximum.

Withdrawal force test:

Steel test pin diameter: .0355 +.0000/-.0001 inch.

Insertion depth: .050/.075 inch.

Test pin finish: 16 microinches.

Withdrawal force: 1 ounce minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:01Z
Check the source to verify that this is the current version before use.

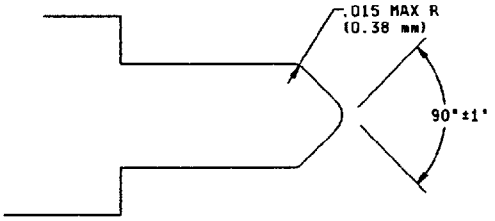


FIGURE 2. Test pin data.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Connectors shall be mounted in mounting hole shown on figure 1 with mating end capped. Test applicable to mounting seal only.

Air pressure: 30 psi.

Duration: 30 seconds minimum, 2 minutes maximum.

Insulation resistance: Method 3003, MIL-STD-1344; 5,000 megohms minimum.

Center contact retention:

Axial force: 6.0 pounds minimum.

Radial torque: 4 inch-ounces minimum.

Solderability 1/: Method 208 of MIL-STD-202.

Resistance to test probe damage: Not applicable.

Corrosion (salt spray): Method 1001, test condition B, MIL-STD-1344.

Voltage standing wave ratio (VSWR): Not applicable.

Connector durability: Method 2016 of MIL-STD-1344; 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance (in milliohms maximum):

	Initial	After environment
Center contact:	3.0	4.0
Outer contact:	2.0	Not applicable

1/ For quality conformance inspection, the test shall be performed in group B following the insulation resistance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:01Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage at sea level: Method 3001 of MIL-STD-1344; 1,000 volts rms.

Vibration, high frequency: Method 2005, test condition IV, MIL-STD-1344.

Shock: Method 2004, test condition G, MIL-STD-1344.

Thermal shock: Method 1003, test condition A, MIL-STD-1344 except test high temperature shall be +200°C.

Humidity: Method 1002, type II, of MIL-STD-1344. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 250 volts minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 670 volts rms at 5 MHz.

Leakage current: Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: Not applicable.

RF insertion loss: Not applicable.

Group qualification: See table I.

TABLE I. Group qualification and retention testing.

Group	PINs 1/ 2/
I	M39012/60-3001 M39012/60-3002 M39012/61-3001 M39012/61-3002
II	M39012/60-4001 M39012/60-4002 M39012/61-4001 M39012/61-4002

1/ Submission and qualification of any connector in a group qualifies any and all connectors of the entire group.

2/ For qualification retention, data may be supplied on any part in a group to retain qualification for that entire group.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:01Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AM, AT, ME, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - ES

(Project 5935-3911-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:01Z
Check the source to verify that this is the current version before use.